

Hoang Duc Viet

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SUMMARY

Former AI leader at an SEO agency in Hanoi, Vietnam, where I led a team to deliver production AI systems spanning AI Agents, LLM Workflows, and Machine Learning / Deep Learning solutions. I'm now looking for ambitious applications of AI in exciting niches.

EXPERIENCE

SEONGON Joint Stock Company

Hanoi, Vietnam

Search Marketing (SEO and Google Ads) agency

AI Leader

May 2025 – Aug 2026

- Founded and led an AI team of 5 to deliver AI systems that transformed how the company does search marketing workloads.
- Built a Human-Agent collaboration platform serving 120 people, with over 20 AI solutions spanning AI Agents, LLM Workflows, and several Machine Learning workflows.
- Designed and implemented the platform's AI system standards: Nonhuman Identity, Observability, Evaluation, Human-in-the-loop (HITL), Feedback-as-a-feature, and Cost tracking.
- Delivered a 2x time reduction across the SEO production chain, with tasks like internal linking and content-outline generation taking up to 5 times less time than before, at the same quality as senior staff.
- Trained 80% of the company to use agentic AI (Claude Code), with more than 50 measurable outcomes after training.

Operations Assistant to the CEO

Dec 2024 – May 2025

- Worked directly with the CEO on daily operations and quarterly planning, and ran market-intelligence work that shaped the company's strategic AI moves.
 - Promoted into the AI Leader role after the CEO judged I'd bring more value to the company there.
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SELECTED PROJECTS

Company-wide AI platform

Built on the Vercel AI SDK and serving 120 people at a Search Marketing agency, it hosts over 20 AI solutions spanning AI Agents, LLM Workflows, and Machine Learning workflows: 19 agents, 21 connectors in production.

To keep it reliable and scalable without slowing development, I built the supporting layer:

- **Feedback-as-a-feature:** custom UI widgets after every model trace and key log, so live users can rate and comment on production, with feedback flowing into a shared database.
- **Cost tracking:** a dashboard breaking cost down by department, agent, and person over time, built on the model and non-model traces.
- **Human-Agent Access Control:** strict, task-scoped data access for both people and agents, so each reaches only what its job requires.

Two of its agents are detailed below; many more run in production.

Tech stack: Vercel AI SDK, Next.js, TypeScript, Python, PostgreSQL, Larksuite Open API

Semantic Keyword Clustering

An LLM/ML workflow that groups tens of thousands of Vietnamese keywords into semantic clusters. It uses voyage-3-large (2,048-dim) embeddings, UMAP for dimensionality reduction, and HDBSCAN, with Vietnamese tokenisation (underthesea), k-NN outlier voting, and recursive splitting, then a light-weight LLM to label each cluster. Reduced average keyword clustering time from **1 hour** (depending on the number of keywords) to **5 minutes**.

Tech stack: Python, OpenAI embeddings, UMAP, HDBSCAN, underthesea

Multi-agent System for Making Presentations

An agentic system that turns a client brief and raw Excel data into a finished B2B SEO proposal with editable elements. All user inputs generate a comprehensive outline for human approval and adjustment before generating slides. Reduce average time spent for making an SEO proposal from **1 day** to **1.5 hours**.

- **Orchestrator-Executor pattern:** an orchestrator agent coordinates and dispatches appropriate tasks to sub-agents for optimal context handling with enormous client data. Sub-agents include the Outlining Agent, Slides Agent, SEO Expert Agent, and Data Analyst Agent.
- **Data Analysis:** a data-analyst sub-agent for handling complex, mix-structured data files such as human-created Excel spreadsheets, connected to the E2B sandbox service with strict hooks for smart data handling.
- **Editable Slides and Components:** a library of available components is pre-built (diagrams, primitive shapes, layout). An agent selects the appropriate components to convey the data and passes the correct data schema in JSON to the system. A deterministic rendering engine takes care of linting, such as avoiding overlapping elements and overflowing text.

Tech stack: TypeScript, Vercel AI SDK, E2B, pptxgenjs / resvg / fontkit, PostgreSQL, Cloudflare R2

TECHNICAL SKILLS

Languages	Python, TypeScript, SQL
Machine learning	training neural networks from scratch (PyTorch), LoRA / QLoRA fine-tuning, embeddings, clustering (HDBSCAN), dimensionality reduction (UMAP), scikit-learn, TensorFlow
Agents & LLMs	multi-agent orchestration (Orchestrator-Executor), tool use / function calling, multi-provider routing (model gateway), evaluation (evals), guardrails, human-in-the-loop (HITL), observability, Model Context Protocol (MCP)
Retrieval & search	BM25 / BM25F, PageRank, cross-encoder reranking, retrieval-augmented generation (RAG), hybrid search, vector search (pgvector), web crawling and indexing
Infrastructure	Next.js, FastAPI, PostgreSQL, Redis, Docker, Railway, Vercel AI Gateway, E2B, Larksuite Open API, Tavily

CERTIFICATIONS

Generative AI Leader	<i>Google</i>
Claude Code in Action	<i>Anthropic</i>
Google AI	<i>Google</i>
Google Project Management Professional Certificate	<i>Coursera</i>
Data Analysis with Python	<i>freeCodeCamp</i>

EDUCATION

Foreign Trade University	2023 – 2027 (expected)
B.A., International Business. Hanoi.	
High School for Gifted Students, Hanoi National University of Education	2020 – 2023
English specialization. Two-time runner-up, HNUE Excellent Student Selection Exam in English.	
